Proxmox VE Cluster — Test VM & SDN Decommissioning Runbook

Complete Command Execution Guide, HA Disarmament, Storage Purge & Network Teardown

TARGET CLUSTER	ACTION SCOPE	STORAGE RECLAIMED	PRODUCTION STATE
cisai-pve (prox022, prox023, prox024)	9000-Series VMs & VXLAN SDN	~43 GiB Ceph/ZFS Space Freed	VMs 100, 102, 106, 103, 104 Intact

Operational Purpose: Following completion of datacenter physical resiliency and performance testing, this document records the exact procedure and commands executed to cleanly purge all synthetic 9000-series test VMs, disarm High Availability (HA) manager tracking, delete ZFS storage replication jobs, and permanently decommission the overlay VXLAN Software-Defined Network (SDN).

1. Decommissioned Artifacts Inventory

Purged Resources

Resource Type	Identifier / Name	Resident Node	Underlying Storage / Network	Purge Result
Virtual Machine	VM 9001 (ha-test-ceph-ubuntu2204)	prox022	Ceph RBD (32GB pool ceph)	Stopped & Purged
Virtual Machine	VM 9002 (zfs-test-prox24-ubuntu2204)	prox024	Ceph RBD (32GB pool ceph)	Stopped & Purged
Virtual Machine	VM 9003 (zfs-benchmark-prod-vm)	prox023	ZFS Zvol (32GB pool zfs-ha)	Stopped & Purged
Virtual Machine	VM 9004 (vm-9004-test)	prox023	Ceph RBD (32GB pool ceph)	Stopped & Purged
HA Service	vm:9001, vm:9002, vm:9003, vm:9004	Cluster-wide	pve-ha-crm / pve-ha-lrm	Disarmed & Removed
Replication Jobs	9002-0, 9002-1, 9003-0, 9003-1	Cluster-wide	pvesr storage replication	Deleted
SDN VXLAN Zone	myzone (MTU 1450, tag 10010)	All 3 Nodes	192.168.10.x Full-Mesh Peers	Deleted & Unmounted
SDN VNet & Subnet	vnet10 & myzone-10.10.10.0-24	All 3 Nodes	Kernel bridge & overlay tunnel	Interfaces Deleted

2. Step-by-Step Command Execution Guide

CLI Runbook

Phase 1: Delete Storage Replication Jobs

Removes periodic ZFS volume synchronization schedules from `/etc/pve/replication.cfg` before destroying VMs.

```
# Run on cluster master (prox022):
pvesr delete 9002-0
pvesr delete 9002-1
pvesr delete 9003-0
pvesr delete 9003-1
```

Phase 2: Disarm High Availability (HA) Manager

Removes test VMs from the cluster watchdog CRM. Without this step, killing a VM would trigger an automated HA fence or restart attempt.

```
# Run on cluster master (prox022):
ha-manager remove vm:9001
ha-manager remove vm:9002
ha-manager remove vm:9003
ha-manager remove vm:9004
```

Phase 3: Stop and Purge Test VMs with Storage Disks

Executes forceful shutdown and triggers storage backend garbage collection (`--purge 1`).

```
# 1. On prox022 (hosts VM 9001):
qm stop 9001 --skiplock 1
qm destroy 9001 --purge 1 --skiplock 1

# 2. On prox024 (hosts VM 9002):
qm stop 9002 --skiplock 1
qm destroy 9002 --purge 1 --skiplock 1

# 3. On prox023 (hosts VMs 9003 and 9004):
qm stop 9003 --skiplock 1
qm destroy 9003 --purge 1 --skiplock 1

qm stop 9004 --skiplock 1
qm destroy 9004 --purge 1 --skiplock 1
```

Phase 4: Completely Remove Proxmox SDN (VXLAN)

Tears down cluster SDN definitions, eliminates boot-persistence scripts, and deletes live Linux kernel interfaces.

```
# Step 4.1: Delete cluster SDN configs & IPAM state (run on prox022):
rm -f /etc/pve/sdn/zones.cfg
rm -f /etc/pve/sdn/vnets.cfg
rm -f /etc/pve/sdn/subnets.cfg
rm -f /etc/pve/sdn/pve-ipam-state.json
rm -f /etc/pve/sdn/.running-config

# Step 4.2: Delete network interface persistence files (run on prox022, prox023, prox024):
rm -f /etc/network/interfaces.d/sdn

# Step 4.3: Tear down live kernel interfaces (run on prox022, prox023, prox024):
ip link set vnet10 down 2>/dev/null || true
ip link delete vnet10 2>/dev/null || true
ip link delete vxlan_vnet10 2>/dev/null || true
```

3. Post-Cleanup Verification Commands & Health Audit

Verification Table

Verification Command	Target Validation Check	Expected Output	Live Audit Result
`qm list`	Confirm 9000-series VMs are absent	Only VMs 100, 102, 106, 103, 104	✔ **PASSED (9000s Absent)**
`ha-manager status`	Confirm HA service list is clean	Zero test VM services registered	✔ **PASSED (0 Test Services)**
`pvesr list`	Confirm replication queue	Empty table	✔ **PASSED (No Jobs)**
`ip -br link | grep -E "vnet|vxlan"`	Confirm SDN kernel teardown	Empty output (no links)	✔ **PASSED (Clean)**
`ls -la /etc/pve/sdn`	Confirm cluster SDN directory	Total 0 (clean directory)	✔ **PASSED (Empty)**
`ceph status`	Verify Ceph storage state	HEALTH_OK • 12/12 OSDs up & in	✔ **PASSED (HEALTH_OK)**

4. Surviving Production Workloads (100% Intact)

Active Services

VMID	Virtual Machine Name	Host Node	Allocated Memory	Boot Storage Backend	Operational Status
100	windows	prox022	40,960 MB	local-lvm:vm-100-disk-1	Running (PID 44173)
102	testdontremove	prox022	20,480 MB	local-lvm:vm-102-disk-0	Running (PID 900303)
106	2026090578-jishnua-ubuntu24	prox022	32,768 MB	local-lvm:vm-106-disk-0 (250GB)	Running (PID 1958557)
103	115-gouthamoshibu-ubuntu24	prox024	10,240 MB	local-lvm:vm-103-disk-0 (150GB)	Running (PID 3714007)
104	2026090558-gouthamoshibu-ubuntu24	prox024	10,240 MB	local-lvm:vm-104-disk-0 (150GB)	Running (PID 3776614)